

Difficulties Installing OpenVAS (Greenbone) and Nessus

The first attempt to install the vulnerability scanner was with OpenVAS (Greenbone Vulnerability Manager). This attempt was unsuccessful due to a combination of a slow, unstable network connection and limited disk space on the virtual machine. The feed synchronization process (`greenbone-feed-sync`), which downloads several gigabytes of vulnerability test data (NASL, Notus, SCAP, and CERT-Bund files) via `rsync`, repeatedly failed with connection timeouts and DNS resolution errors. The VM's virtual disk also had insufficient free space (initially under 1 GB available), which required resizing the partition before the installation could proceed.

After this failure, Nessus Essentials was attempted as an alternative scanner. This proved even more difficult: the installation process was affected by a misconfigured Firefox proxy setting (left over from a prior Burp Suite configuration) that blocked access to the activation and download servers, as well as repeated network interface failures inside the Kali VM (the virtual network adapter losing its “carrier” connection, and DNS resolution breaking intermittently). Troubleshooting Nessus consumed approximately three hours without reaching a working, activated installation.

Following this, the installation effort returned to OpenVAS/Greenbone. With the network connectivity issues diagnosed and resolved (switching the VM's network adapter to NAT mode for the duration of the download, and fixing DNS configuration), the feed synchronization was restarted and allowed to run to completion over an extended period due to the slow connection speed. This second attempt succeeded, and `gvm-check-setup` confirmed a complete installation with all NVT, SCAP, and CERT data present.

Difficulty Configuring the Server/Scanner Computer

After installation, further difficulties arose in getting the GVM services (`gvmd`, `gsad`, `osspd-openvas`) to start and remain stable. The `gvmd` service required an extended first-run period to import the newly downloaded vulnerability data into its database, during which login attempts to the web interface failed with authentication errors. The `gsad` web service also produced intermittent port-binding conflicts on startup, requiring manual process cleanup and service restarts before the web interface became reliably accessible.

Difficulty Placing the Machine on the Shared Lab Network

Placing PC 5 onto the same isolated lab network (192.168.50.0/24) as the other lab machines also presented challenges. The VirtualBox network adapter had to be switched between NAT mode (required temporarily for internet access during the scanner installation) and Bridged mode (required for connectivity to the other lab VMs over the shared network). Each switch required reconfiguring the VM's network interface, and at several points the virtual network adapter failed to establish a link (“no carrier”) after being reattached, requiring the adapter's cable connection to be manually toggled in VirtualBox before the interface would come back online.